

Final Report - Town of Forests

Wildfire Risk Modeling Exercise - WIRC/SJSU

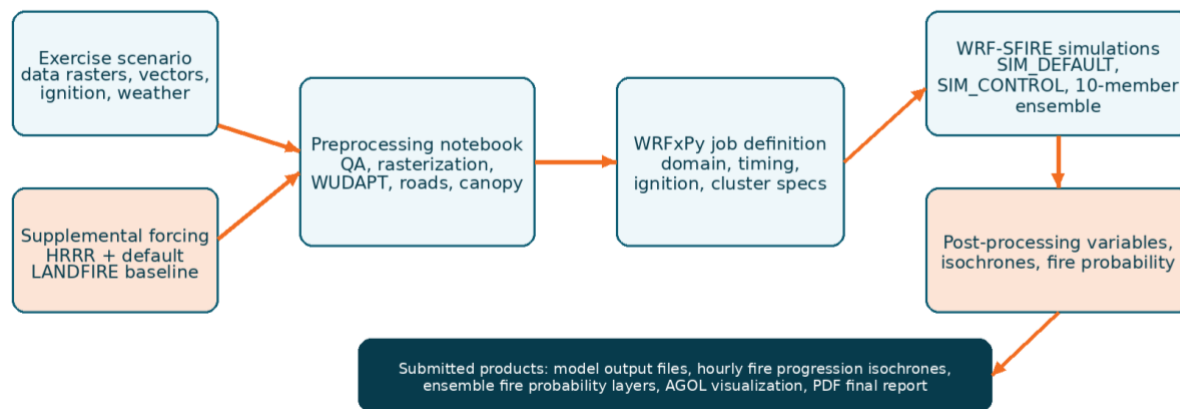
Scenario: Module A - Town of Forests

Model: WRF-SFIRE coupled atmosphere-fire simulations automated with WRFxPy

Ignition: [38.9014, -120.0306] on 2024-10-27 at 1700 UTC

Scenario setting: Mountainous terrain, variable elevation, high tree density, and residential subdivisions in the wildland-urban intermix.

A. Workflow Diagram + Full Workflow Description

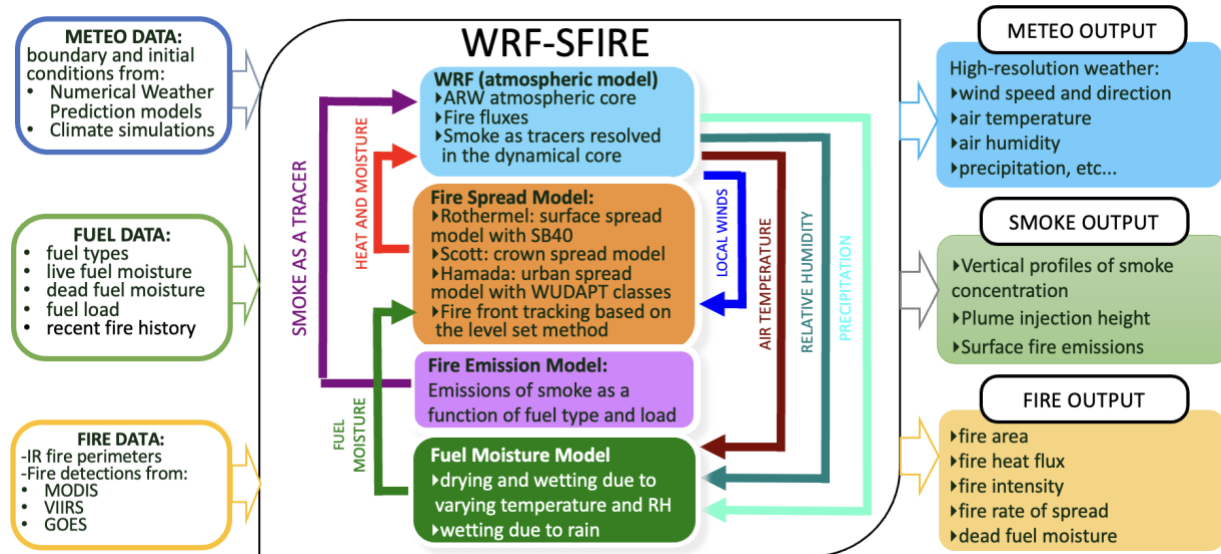


Submission: The workflow uses the exercise ignition, fuels, terrain, trees, buildings, roads, and weather data for Town of Forests. Data were prepared in `preproc/data_preparation_forest.ipynb`, then run through WRFxPy and WRF-SFIRE. The final products are model-variable files, hourly fire progression isochrones, and an ensemble fire probability layer.

Workflow narrative

1. Access scenario data from the Wildfire Commons workspace, including rasters, GeoJSON vectors, ignition data, tree lists, building attributes, road networks, and synoptic weather data.
2. Run the scenario-specific preprocessing notebook to read ignition data, visualize weather, inspect fuels and terrain, and check consistency between provided fuel properties and WRF-SFIRE namelist options.
3. Preprocess built-environment data by rasterizing building polygons into building side length, separation length, and height; classify urban areas into WUDAPT classes; rasterize roads and interpolate them into dominant fuel classes.
4. Preprocess vegetation data by rasterizing tree properties into canopy-top, canopy-bottom, and canopy-bulk-density layers compatible with the WRF-SFIRE crown-fire configuration.
5. Create the WRFxPy job JSON for the scenario, including simulation start/end timing, HRRR atmospheric forcing, domain configuration, ignition information, requested post-processing variables, and cluster/HPC configuration.
6. Run deterministic simulations for SIM_BASELINE and SIM_CONTROL and generate the 10-member ensemble by perturbing fuel load, fuel moisture, and horizontal wind components.
7. Post-process model outputs to retain FXLONG, FXLAT, NFUEL_CAT, and FIRE_AREA and derive hourly fire progression isochrones and the ensemble fire probability map.
8. Prepare non-proprietary output layers and AGOL-compatible map/feature services for evaluator review; add code/notebook/output links to Wildfire Commons Additional Resources.

B. Model Architecture Diagram + Explanation



Submission: The model is a dynamic coupled atmosphere-fire spread model. It uses WRF for atmospheric forcing and SFIRE for fire spread, with exercise-specific fuel, canopy, road, and urban inputs.

- Major components: WRF atmosphere, SFIRE level-set fire spread, crown fire using modified Scott (2001), WUDAPT urban fuel classes, HAMADA urban rate of spread, and RK3-WENO5 level set with reinitialization.
- Inputs: exercise data (GeoTIFF rasters and GeoJSON files), derived fuel/canopy/urban layers, HRRR forcing, and WRFxPy job JSON file (forest.json).
- Outputs: burned-area/model variables, hourly fire isochrones, and ensemble fire probability.
- Hardware: CPU/HPC workflow run through WRFxPy; no GPU required. Node and processor settings are stored in the WRFxPy job JSON. Each WRF-SFIRE simulation used 42 cores (7 x 6 parallel patches) and 2 ensemble members were run in a single ARM-based Grace-Grace node with 144 cores.
- Design choice: SIM_BASELINE provides a LANDFIRE-only and equilibrium fuel moisture baseline; SIM_CONTROL uses exercise-aligned fuels, moisture, buildings, roads, and canopy; SIM_ENSEMBLE samples uncertainty in fuel load, fuel moisture, and wind inputs.

C. Inputs Used (Core + Derived / Model-Relevant)

Submission: All relevant core inputs were used. Core rasters and GeoJSONs are in WGS84 / EPSG:4326 unless otherwise stated. The scenario extent is approximately 6 km x 5 km. All inputs are visualized in the data preparation Jupyter notebook.

- Forest Ignition (.geojson): ignition point and start time; used to initialize the simulation. [38.9014, -120.0306] on 2024-10-27 at 1700 UTC
- Forest Weather (CSV): timestamp, temperature, relative humidity, wind speed, and wind direction; used for scenario comparison and to match the HRRR forcing time.
- Forest Surface Fuels (.tif): SB40, depth, moisture, rhof1, rhof10, rhof100, and SAV; used to construct WRF-SFIRE fuel and fuel-moisture fields.
- Forest Elevation (.tif): terrain input used by the model.
- Forest Trees (.geojson): canopy height (HT), canopy base height (CBH), and canopy bulk density (CBD); used to derive canopy layers for crown fire model.
- Forest Buildings and Roads (.geojson): buildings were rasterized into side length, separation length, and height; roads were rasterized and used to adjust/interpolate fuel classes.

- Derived fuel layer: combines SB40, WUDAPT urban classes, and road interpolation; used as the WRF-SFIRE fuel input.
- Derived canopy layers: canopy top, canopy bottom, and canopy bulk density; derived from tree properties and used by the crown fire model.
- HRRR forcing: gridded atmospheric initial/boundary conditions matched to the scenario date/time because WRF-SFIRE requires gridded weather forcing. For this case, HRRR winds were similar to the scenario wind magnitude/direction; the main differences were temperature and relative humidity.
- WRF-SFIRE inputs: namelist files and input and boundary NetCDF files wrfinput and wrfbdy are available in the folder wrfsfire_inputs.
- Ensemble perturbation fields: fuel load (fgi), fuel moisture (fmc), wind components (u_wind and v_wind); used to create the 10-member ensemble.

D. Model Outputs

Submission: The primary outputs describe fire progression and burn probability under the scenario ignition and forcing. Outputs are intended for map review and comparison across deterministic and ensemble runs.

- Model files - folder: data/output/model. Format/data type: WRF/NetCDF model output. Variables retained: FXLONG, FXLAT, NFUEL_CAT, FIRE_AREA. Cases: SIM_BASELINE, SIM_CONTROL, SIM_ENSEMBLE.
- Hourly fire progression isochrones - folder: data/output/agol. Format/data type: geospatial vector layers for 50% of the ensemble members. Interpretation: predicted fire spread timing by hour.
- Fire probability map - folder: data/output/agol. Format/data type: geospatial probability layer from 10 ensemble members. Values represent the fraction of ensemble members that burned each location, from 0% to 100% in 10% increments.
- Spatial reference: EPSG:4326 for submitted geospatial layers.
- AGOL feature/map service links:
<https://ulri.maps.arcgis.com/apps/mapviewer/index.html?webmap=a3c12ec224b4461387fb8277032b6daf>

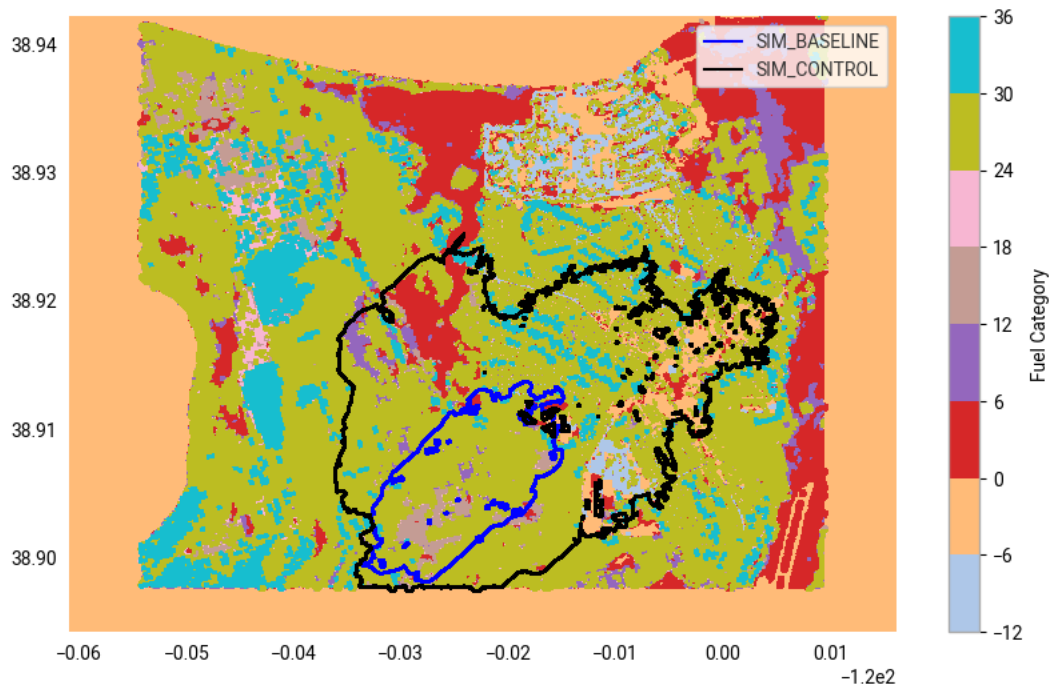


Figure 1: Fuel classification layer combining SB40, WUDAPT and road mask and comparison of the SIM_BASELINE and SIM_CONTROL. Note that differences are driven mainly by changes in fuel moisture (much drier in the SIM_CONTROL) and adding urban capabilities (unburnable for SIM_BASELINE).

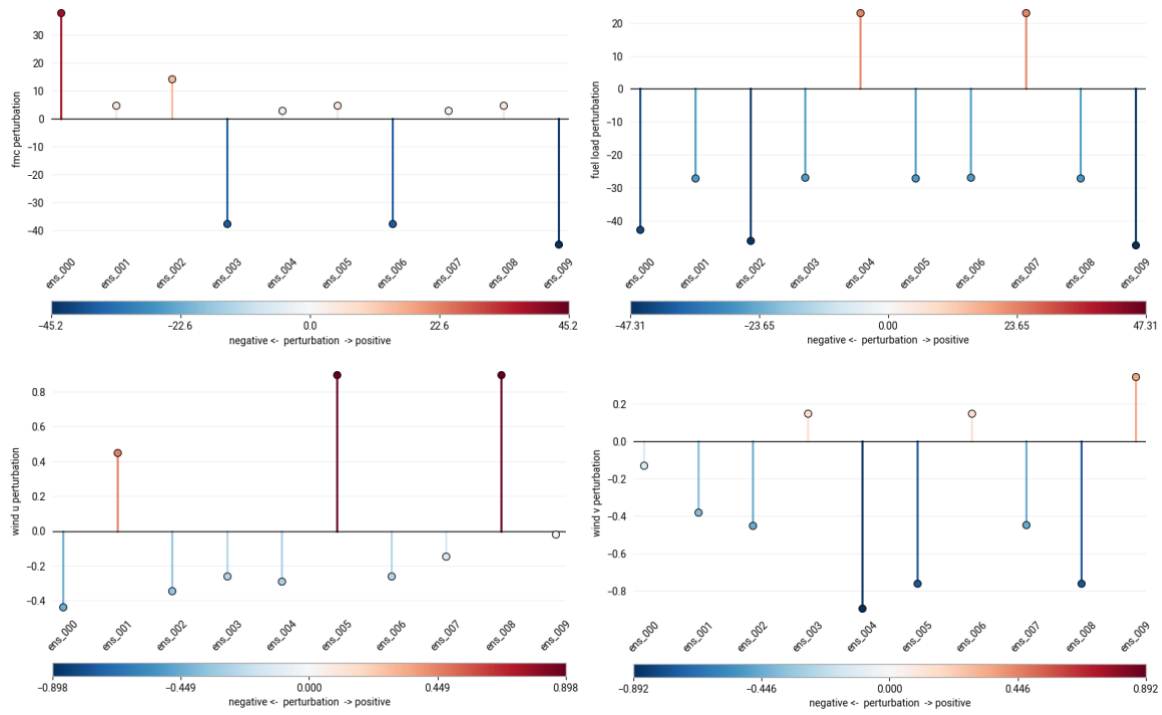


Figure 2: Random ensemble perturbations for the 10-member ensemble (SIM_ENSEMBLE) to the critical input variables to account for their uncertainties. Relative perturbations of fuel moisture (fmc) and fuel load (fui) in percentage, and absolute perturbations to the wind components (wind u) and (wind v) in m/s.

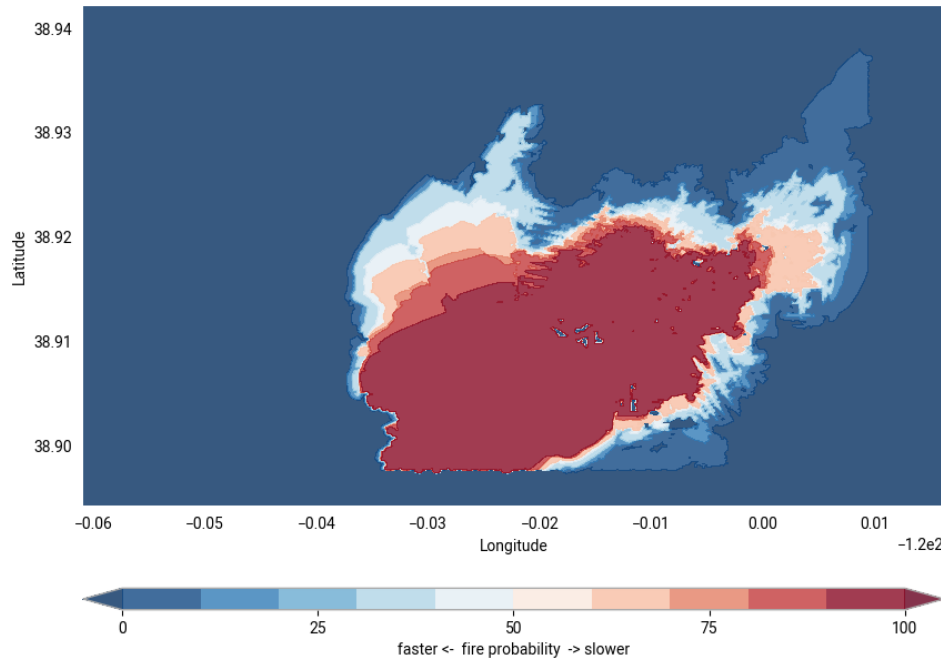


Figure 3: Fire probability map generated from the WRF-SFIRE 10-member ensemble SIM_ENSEMBLE.

E. Downstream Use of Model

Submission: The main use is response resource planning during initial attack. The hourly isochrones and probability map can also support incident size-up, readiness monitoring, pre-incident planning, and community wildfire risk or mitigation prioritization.

- For Town of Forests, the outputs help identify where the simulated fire may move, when it may reach WUI/intermix areas, and where ensemble uncertainty is highest.

F. Assumptions, Parameters, and Validation

Submission: Key assumptions are that no suppression is modeled, core synthetic data are retained where possible, and buildings/roads affect the fire primarily through derived fuel and urban-class inputs rather than through a separate structure-loss model.

- Weather: HRRR forecast forcing was substituted for surface observations to satisfy WRF-SFIRE requirements while matching the scenario day/time. For this case, HRRR winds were similar to the scenario wind magnitude/direction; the main differences were temperature and relative humidity.
- Fuel moisture: SIM_BASELINE initializes from equilibria and runs the time-lag fuel-moisture model; SIM_CONTROL and SIM_ENSEMBLE uses the exercise fuel moisture and keeps it constant.
- Ensemble size: 10 members because of computational resources.
- Fuel uncertainty: fuel load (fgi) and fuel moisture (fmc) perturbations use 30% standard deviation and are bounded from -50% percent to +50% percent.
- Wind uncertainty: wind components (u_wind and v_wind) perturbations use 0.5 m/s standard deviation and are bounded from -2.0 m/s to +2.0 m/s.
- Validation/QA: inputs were checked by visualizing weather/fuel/terrain fields, comparing weather with station data, checking fuel properties against WRF-SFIRE namelist options, and reviewing deterministic and ensemble output layers. No observed fire perimeter was available for outcome validation.

G. Additional Supporting Documentation

Submission: The workflow and model files are documented in the public submission repository and related model documentation.

- Submission repository: https://github.com/wirc-sjsu/Fire_Risk_Exercise
- README: repository README.md
- Data preparation notebook: preproc/data_preparation_forest.ipynb
- WRFxPy job file: wrfxpy_jobs/forest.json
- Model documentation: WRF-SFIRE - <https://github.com/openwfm/WRF-SFIRE>; WRFxPy - <https://github.com/openwfm/WRFxPy>
- Output scripts and notebook: inside postproc folder
- Output folders: outputs/model and data/output/model and data/output/agol
- AGOL Web Map with final results:
<https://ulri.maps.arcgis.com/apps/mapviewer/index.html?webmap=a3c12ec224b4461387fb8277032b6daf>